

SOLVOFIN URBAN INTELLIGENCE

MUNICIPAL TRANSIT COMPUTER VISION & ROAD SAFETY AUDIT REPORT

Report ID: REP-00101 | Generated: 2026-09-19 07:22:29

1. MEDIA & LOCATION AUDIT METRICS

Filename	transit_route_18_coastal_highway.mp4	Media Type	VIDEO
Media ID	MEDIA-00101	File Size	27.08 MB
Uploaded By	Fleet Operator #42	Analysis Status	COMPLETED
Upload Location	17.7289, 83.3184 (±6.5m)	Transit Bus Route	BUS-18-NORTH
Scene Location	17.7342, 83.3248	AI Model Version	SOLVOFIN-GEMINI-3.7-FLASH-URBAN

2. EXECUTIVE SUMMARY & ROAD HEALTH SCORE

Comprehensive audit of Coastal Corridor NH-16 (Route 18). Detected 4 road defects including 2 severe potholes. Tracked 48 unique vehicles with flow rate of 64 veh/min. 2 commercial vehicles flagged for visible dark exhaust smoke. 1 pedestrian risk incident recorded at faded crosswalk.

Road Health Score	58/100 (MODERATE)	Potholes Detected	2
Surface Distress Score	42/100	Total Road Defects	5
Defect Density	2.8 defects/km	Signage Condition	65/100

3. 9-POINT MANDATORY ROAD DEFECT & INFRASTRUCTURE AUDIT (YES / NO)

AUDIT VERDICT: 4 OF 9 ROAD HAZARD PARAMETERS DETECTED (YES) — MUNICIPAL REMEDIATION DISPATCH REQUIRED

Item & Hazard Class	Present ?	Count / Extent	Confidence	Severity	Statutory Civil Action Required
1. Potholes / Deep Cavities	YES	2 detected	94%	HIGH	Immediate VG-30 Bitumen infill & rolling compaction within 24h SLA
2. Alligator Fatigue Cracking	NO	0 (Clear)	99%	NOMINAL	Full-depth pavement reclamation and structural asphalt overlay
3. Longitudinal / Transverse Crack	YES	1 detected	88%	MEDIUM	Polymer elastomeric hot-pour crack sealant to prevent water infiltration
4. Monsoon Waterlogging / Ponding	NO	0 (Clear)	99%	NOMINAL	De-silt road storm conduits and install camber run-off drainage grating
5. Open Manhole / Drain Grate	NO	0 (Clear)	99%	NOMINAL	Emergency 2-hour SLA barricade placement & heavy-duty ductile iron cover
6. Pavement Rutting / Depression	NO	0 (Clear)	99%	NOMINAL	Cold milling leveling followed by high-stability bitumen binder course
7. Faded Zebra / Lane Marking	YES	1 detected	91%	HIGH	Thermoplastic retroreflective paint application with glass bead embedment
8. Missing / Broken Signboard	YES	1 detected	87%	LOW	Fabricate & erect high-intensity prismatic retroreflective road sign
9. Damaged Median / Divider	NO	0 (Clear)	99%	NOMINAL	Structural barrier realignment, precast concrete replacement & hazard reflectors

4. INDIVIDUAL ROAD DEFECTS & CAVITY LOG

Defect ID	Type	Severity	Dimensions	Asphalt	Est. Cost	Priority	Work Order
DEF-01	POTHOLE	HIGH	80x65x12cm	0.28 T	Rs. 8,500	90/100	WO-GEN
DEF-02	CRACK	MEDIUM	240x15x4cm	0.12 T	Rs. 4,200	65/100	WO-GEN
DEF-03	FADED ZEBRA CROSSING	HIGH	Surface Area	0.00 T	Rs. 3,500	50/100	WO-GEN
DEF-04	DAMAGED SIGNBOARD	LOW	Surface Area	0.00 T	Rs. 3,500	50/100	WO-GEN
DEF-05	POTHOLE	HIGH	80x65x12cm	0.28 T	Rs. 8,500	90/100	WO-GEN

5. VEHICLES, TRAFFIC, ANPR & VISIBLE EMISSIONS

Total Unique Vehicles	5	Traffic Flow Rate	64 veh/min
Traffic Density	MODERATE	Congestion Index	46/100 (LIGHT)
Visible Smoke Events	2	People Visible (Aggregate)	18 (M: 11, F: 7)*
Visible Buildings Tallied	4	Safety Incidents	2

ANPR Vehicle Registration Index

Track ID	License Plate	OCR Confidence	Status	Jurisdiction
CAR-101	AP39TG2041	96.8%	VERIFIED	Andhra Pradesh
BUS-104	AP31Z9884	94.5%	VERIFIED	AP State Transit
TRUCK-108	AP39AB1234	96.4%	VERIFIED	Andhra Pradesh Commercial
AUTO-112	AP31TA5512	89.2%	VERIFIED	Andhra Pradesh
MOTO-118	AP39BK9021	62.0%	LOW CONFIDENCE	LOW CONFIDENCE (Partial OCR: AP39BK???)

* Notice: Road Health Scores & apparent demographic estimates are AI-derived computational indicators, not official engineering certifications or biometric tracking. L

AI-ASSISTED DECISION SUPPORT & AUDIT APPENDIX

Grounded Telemetry Synthesis • Authoritative RAG Evidence • Human-in-the-Loop Verification Audit

Case / Entity ID: MEDIA-00101 | Generated: 2026-09-19 07:22:29 UTC

AI-GENERATED SUMMARY (ADVISORY OVERVIEW)

Finding: POTHOLE

Context: Coastal Corridor NH-16 Sector 4 • 2026-09-01T06:32:14.000Z • Surface distress index: 42/100 across surveyed lane segment.

1. OBSERVED TELEMETRY VS. AI INTERPRETATION

Dimension / Parameter	Verified Observed Value / AI Interpretation
Media Asset ID	MEDIA-00101
Filename	transit_route_18_coastal_highway.mp4
Upload Date	2026-09-01T06:32:14.000Z
Road Health Score	58/100
Road Surface Condition	MODERATE
Potholes Cataloged	2
Total Infrastructure Defects	5
GPS Location	17.73420, 83.32480
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AI Interpretation (Advisory)	Moderate maintenance priority. The condition compromises commuter comfort or material service life, but does not present an immediate catastrophic hazard.
Suggested Attention Level	MODERATE
AI Decision Rationale	Moderate maintenance priority. The condition compromises commuter comfort or material service life, but does not present an immediate catastrophic hazard.
AI-Assisted Recommendation	Perform physical inspection of the flagged component during the upcoming scheduled maintenance cycle.
Target Action Group	Road Works Maintenance Crew

2. KNOWLEDGE-BASE EVIDENCE (DOCUMENT RAG PROVENANCE)

Organization	Authoritative Source	Section / Page	Verifiable Excerpt
United Nations Economic Commission for Europe (UNECE)	UNECE Regulation No. 107 - Uniform Provisions Concerning Public Service Vehicles (Buses)	Annex 3: Passenger Compartment Interior Safety, Handrails, Emergency Exits and Gangways (p. 42-51)	To ensure passenger survivability and prevent injuries during transit operations, public service buses (Category M2 and M3) must comply with...
Automotive Research Association of India (ARAI) & MoRTH	AIS-140: Intelligent Transportation Systems (ITS) - Requirements for Public Transport Vehicles	Part 1, Section 4 & 5: Vehicle Location Tracking, Driver Vigilance Advisory Integration, and Emergency Escalations (p. 14-22)	AIS-140 specifies safety and telematics architectures for public transport vehicles (buses and commercial passenger fleets): 1. Vehicle Loca...
United Nations Human Settlements Programme (UN-Habitat)	United Nations SDG 11: Sustainable Cities and Communities - Target 11.2 Framework	Target 11.2: Safe, Affordable, Accessible and Sustainable Transport Systems for All (p. Clause 11.2.1)	Target 11.2 of the 2030 Agenda for Sustainable Development calls upon municipal authorities to: "By 2030, provide access to safe, affordable...

3. HUMAN REVIEW & APPEND-ONLY AUDIT TRAIL

Review Dimension	Official Human Oversight Record
Human Review Status	PENDING
Reviewer Identity	Pending supervisor action
Reviewer Role	N/A
Review Timestamp	Pending review
Original AI Recommendation	Perform physical inspection of the flagged component during the upcoming scheduled maintenance cycle.
Modified Recommendation	None (Accepted as original or pending)
Reviewer Comments	No reviewer comments logged.

4. RESPONSIBLE AI DISCLOSURES, DATA GAPS & SDG 11 ALIGNMENT

Human Oversight	Work orders generated from AI volumetric approximations must be verified by the zonal executive engineer.
Liability Boundary	Contractual liability and contractor billing remain governed by physical field measurement certifications.
Privacy Safeguards	Pedestrian and vehicle imagery are processed under edge masking safeguards.
Missing Data / Gaps	Fleet vehicle registry cross-reference unavailable (bus number not matched to active campus roster). • Sub-surface fastener torque, tensile stress, and material fatigue are not measurable from 2D optical pixels.
System Limitations	Decision Support Advisory Only: This AI analysis provides automated prioritization guidance for human inspectors and does not constitute statutory engineering certification. • Human Review Required: All consequential maintenance actions, bus route suspensions, or work-order authorizations require physical sign-off by a certified municipal transport engineer. • Optical Sensor Constraints: Visual CV inferences can be affected by ambient cabin lighting, occlusions, glare, and camera angle variations. • Document Grounding Scope: Retrieved knowledge is grounded in indexed regulatory frameworks (SDG 11 / UNECE / IRC) and must be cross-checked against current depot SOPs.
SDG 11 Alignment	Solvofin is designed to support safer and more sustainable urban mobility, aligned with SDG 11, particularly the objective concerning safe, accessible and sustainable transport.
Metric Validation	No official UN certification or custom emission metric is claimed; operational indicators reflect actual database sensor logs.

* SOLVOFIN RESPONSIBLE AI: AI recommendations are advisory decision support tools and do not constitute statutory engineering certifications or legal determinations. Strict non-fabrication e